

ATLIQ MEDIA · CODEBASICS RPC #21

Tamil Nadu Assembly Election 2026

AtliQ Media — Neutral Election Briefing

HACKATHON EXECUTION PLAN

ECI public data only · 234 constituencies · 2021 vs 2026

Generated: May 28, 2026 · Descriptive analysis · No predictions, no causal claims

2 · HACKATHON EXECUTION PLAN

Hackathon Execution Plan

TN Election RPC #21 — Master Execution Plan

Challenge: Codebasics Resume Project Challenge #21 — Decoding the 2026 Tamil Nadu Assembly Election

Role: Freelance data analyst for **AtliQ Media** (neutral, ECI-only TV storytelling)

Deadline: 28 May 2026, 11:59 PM IST

Prize: ₹10,000

Primary skill tested: Storytelling (70 pts) > analytical hygiene (30 pts)

This document is the **single source of truth** for execution. Keep it updated as decisions are made (headline, party mapping, turnout source).

0. Recommended winning strategy (data-informed)

0.1 What the starter data already shows

Quick profiling of the provided CSVs (join on `ac_number`):

Signal	2021	2026	Story use
Constituencies	234	234	Full coverage
ACs where winning party label changed	—	163 (~70%)	Q2 Flips — dominant narrative
TVK won (label TVK)	0	108 seats	Q3 Vote share / new entrant
Avg winning margin (% of valid votes)	11.8%	7.7%	Q6 — more fragmented races
Winner vote share >50%	(compute)	14	Q6 — few landslides
Winner vote share <35%	(compute)	61	Q6 — many pluralities
Turnout in CSV	234 ACs filled	0 (blank)	Q5 requires ECI scrape
Flips by region (2021 region tag)	—	South 38, Chennai 30, Kongu/Central/North ~26–27, Delta 15	Q1 Geographic

0.2 Recommended story package (best ROI for rubric)

Headline (example — refine after normalization):

"In 2026, Tamil Nadu's assembly became more fragmented: TVK took 108 seats, seven in ten constituencies changed hands, and average winning margins fell sharply versus 2021."

Pick these 3 research questions:

1. **Q6 — Margin of victory** (fragmentation, landslides vs pluralities)
2. **Q2 — Flip story** (163 seat changes; Sankey or alluvial)
3. **Q3 — Vote share / TVK** (where TVK won; vote share shifts by region)

Optional 4th thread for deck (not a 4th "question"): Q1 geographic — one regional heatmap/small multiples slide.

Do not lead with Q5 (turnout) unless you complete ECI turnout scrape early; it is extra work and not required for a winning story.

0.4 Deep electoral-science layer (extra credit, neutral)

In addition to the six research questions, we surface structural indices from comparative-politics literature on a dedicated **/deep** dashboard page and three deck slides (14–16):

- **Effective Number of Parties** (Laakso–Taagepera, 1979): vote-share weighted fragmentation

- **Pedersen volatility** (1979): net vote-share churn between elections
- **Gallagher LSq** (1991): seat-vote disproportionality under first-past-the-post
- **Representation gap** per party (seat share – vote share)
- **Anti-incumbency rate** by region / reserved category / party
- **Race competitiveness** buckets (multi-cornered / three-way / two-party)
- **District churn** with "full sweep" flag

All measures are descriptive, ECI-only, and add depth without changing the lead narrative. See [docs/METRIC_DEFINITIONS.md](#) for formulas and constitutional context.

0.3 Why this beats other combinations

- Hits **surprise + scale** reviewers want (108 TVK seats, 163 flips, margin collapse).
- Three charts that **look different** (distribution, flow, regional bars/maps).
- Fully **descriptive** — no causal claims about why DMK/AIADMK/TVK gained/lost.
- Avoids reserved-seat controversy (Q4) unless you have a crisp, neutral angle.

1. Non-negotiable rules (disqualification if broken)

- ☐ **Data:** ECI public data only (+ Codebasics starter CSVs). No exit polls, news, social media, opinion.
- ☐ **Tone:** No praise/criticism of party, leader, community, religion, region.
- ☐ **Claims:** No "because" explanations for outcomes; no predictions.
- ☐ **Visuals:** No leader photos; no party symbols/logos.
- ☐ **AI:** Allowed for scrape/clean/chart; you own story selection and neutral framing.

2. Deliverables checklist (strict priority)

Priority	Deliverable	Spec	Weight focus
1	Video	5–7 min, on camera, pitch to AtliQ content head; hook → 2–3 stories → editorial rec → limitations	20 pts storytelling + 20 video
2	Deck	8–10 slides: headline → stories → editorial recommendation → limitations	15 pts
3	GitHub	Public repo; Python/SQL notebook runs end-to-end; README with sources + reproduction	20 pts accuracy
4	Dashboard	Power BI / Tableau / Streamlit (optional)	10 pts (skip-scored if absent)
5	LinkedIn + RPC	Post with mandatory tags; submit post URL on RPC page	Required for evaluation

2.1 Editorial recommendation (deck + video)

Must answer for AtliQ: **What should the 60-minute show cover, in what order, and why?**

Example structure (neutral):

- Segment A (10 min): Statewide seat arithmetic — TVK entry, fragmentation metrics
- Segment B (15 min): Regional map — where flips clustered
- Segment C (10 min): Close races — margins under X%
- Segment D (5 min): Data caveats — live results vs Form-20, turnout sourcing

3. Repository structure (current)

```

Codebasics-Hackathon/
├── README.md
├── HACKATHON_EXECUTION_PLAN.md
├── config/election.json      # Jurisdiction metadata (+ /api/meta)
├── data/
│   ├── raw/                 # tn_*_results.csv, constituency_master.csv
│   ├── external/            # Optional scraped 2026 turnout
│   └── processed/           # ac_comparison.csv, sankey_edges.csv, ...
├── notebooks/
│   └── 01_build_and_explore.ipynb
├── src/                     # metrics, advanced_metrics, party_normalize, charts,
presentation
├── api/                     # FastAPI (serves processed CSVs + /api/advanced/* +
optional web/dist)
├── web/                     # React + TypeScript + ECharts (primary dashboard, incl.
Deep insights page)
├── scripts/                 # build_processed_data, build_presentation,
fetch_eci_turnout_2026, ngrok, ...
├── outputs/
│   ├── charts/              # Plotly HTML exports
│   ├── presentation_assets/  # PNGs embedded in deck
│   └── walkthrough/         # Video script, teleprompter, checklist
├── deck/                   # TN_2026_AtliQ_Briefing.pptx
├── dashboard/              # Legacy Streamlit (optional; not submission path)
└── docs/
    ├── METRIC_DEFINITIONS.md
    ├── DASHBOARD_QA.md
    ├── DASHBOARD_REUSE.md
    └── VIDEO_NARRATION_SCRIPT.md
  
```

4. Data dictionary & joins

4.1 Files

File	Rows	Key
<code>tn_2021_results.csv</code>	~4,232	<code>ac_number</code>
<code>tn_2026_results.csv</code>	~4,257	<code>ac_number</code>
<code>constituency_master.csv</code>	234	<code>ac_number</code>

4.2 Columns (results files)

- `constituency`, `ac_number`, `candidate`, `party`, `votes`, `turnout`, `reserved`, `region`
- Turnout:** populated in 2021; **empty in 2026** → scrape from `results.eci.gov.in/ResultAcGenMay2026` if needed

- **Name drift:** e.g. `Gummidipundi` vs `Gummidipoondi` — always join on `ac_number`, not name

4.3 Regions (editorial, not ECI)

Chennai Metro, North, Central, Kongu, Delta, South — use `constituency_master.region` as canonical.

4.4 Reserved seats

GEN 188 | SC 44 | ST 2 — same in 2021 and 2026.

5. Metric definitions (document in `docs/METRIC_DEFINITIONS.md`)

5.1 Valid votes per AC

```
valid_votes(ac) = SUM(votes) WHERE party ≠ 'NOTA'
```

(Confirm against ECI constituency PDF for 1–2 sample ACs; note definition in README.)

5.2 Winner

```
winner(ac) = candidate with MAX(votes) among non-NOTA candidates
```

Ties: flag manually (should be rare).

5.3 Vote share

```
vote_share(candidate) = votes / valid_votes(ac) * 100
winner_share(ac) = votes(winner) / valid_votes(ac) * 100
```

5.4 Margin of victory

```
margin_votes = winner_votes - runner_up_votes
margin_pct = margin_votes / valid_votes(ac) * 100
```

5.5 Flip

```
flip(ac) = winner_party_2026 ≠ winner_party_2021
```

After **party normalization** (Section 6). Report both raw-label flips (163) and normalized flips in limitations if they differ.

5.6 Seat counts by party

Count ACs where normalized winner party = X.

For “alliance” views, map parties to **formation** (DMK+, AIADMK+, etc.) only if you publish an explicit, source-backed mapping table.

6. Party normalization (critical for 2026)

2026 `party` field has long names and many `IND` rows. Build `party_normalize.csv`:

raw_party (examples)	normalized	notes
TVK	TVK	
DMK	DMK	
Anna Puratchi Thalaivar Amma Dravida Munnetra Kazhagam	AIADMK	verify spelling variants
All India Puratchi Thalaivar Makkal Munnettra Kazhagam	AIADMK	
INC	INC	
BJP	BJP	
NTK	NTK	
IND	IND	do not merge into majors without ECI “party” field proof

Process:

- `SELECT DISTINCT party, COUNT(*) FROM tn_2026 GROUP BY party ORDER BY count DESC`
- Manually map top 30 labels
- For winner stats, use **normalized** column only
- In README: “Independent candidates are labeled IND; vote-transfer claims are not made.”

7. Phase-by-phase execution (step-by-step)

PHASE 1 — Foundation (Days 1–2)

Step 1.1 — Fork starter data

- ☐ Copy `input_files_for_participants_rpc/data/*` → `data/raw/`
- ☐ Never edit raw files; all transforms write to `data/processed/`

Step 1.2 — Create GitHub repo

- ☐ Public repository
- ☐ Add `.gitignore` (exclude large caches, keep processed CSVs <10MB or use Git LFS if needed)
- ☐ Initial README: challenge name, disclaimer, folder map

Step 1.3 — Environment

- ☐ Python 3.11+ with `pandas`, `matplotlib` or `plotly`, `jupyter`
- ☐ Optional: `requests`, `beautifulsoup4` for turnout scrape
- ☐ `requirements.txt` pinned versions

Step 1.4 — Build AC-level tables

Notebook `01_build_ac_tables.ipynb`:

1. Load 2021 + 2026 + master
2. Per `ac_number`, compute: `valid_votes`, `winner`, `runner_up`, `winner_share`, `margin_pct`, `total_candidates`
3. Join master → `district`, `region`, `reserved`
4. Export `data/processed/ac_summary_2021.csv`, `ac_summary_2026.csv`
5. Export `data/processed/ac_comparison.csv` with columns:
`ac_number, region, reserved, winner_2021, winner_2026, flip, margin_2021, margin_2026, winner_share_2021, winner_share_2026`

Validation gates:

- ☐ 234 rows each year
- ☐ Sum of votes per AC matches manual check for AC 1 and AC 100
- ☐ No null `ac_number`

Step 1.5 — Party normalization

- ☐ Create `src/party_normalize.py` + `data/party_normalize.csv`
- ☐ Re-run AC tables with `winner_party_norm`

PHASE 2 — Story metrics (Days 3–4)

Notebook `02_story_metrics.ipynb`:

Q6 — Margins (do first)

- ☐ Statewide: mean/median margin 2021 vs 2026
- ☐ Count: `winner_share` > 50%, < 35% (both years)
- ☐ Chart: overlaid histogram or boxplot of `margin_pct`
- ☐ Table: top 10 closest races 2026 (neutral titles: "Smallest margin, AC name")

Q2 — Flips

- ☐ Count flips (normalized)
- ☐ Build transition matrix: `winner_2021 → winner_2026` (top 15 flows)
- ☐ Chart: Sankey (plotly) or alluvial — **no party colors that look like flags**; use neutral palette
- ☐ Optional: map flips per region (choropleth if you have GeoJSON; else bar chart by region)

Q3 — TVK / vote share

- ☐ TVK seats won by region
- ☐ Vote share by party statewide (2021 vs 2026) — bar chart
- ☐ Regional small multiples: DMK, AIADMK, TVK share by region
- ☐ **Do not claim** “TVK took votes from X” — say “TVK’s vote share was Y% in region Z; DMK share changed from A to B” (descriptive only)

Q1 — Geographic (one slide)

- ☐ Seats won by major normalized parties per region, 2021 vs 2026 side-by-side stacked bar

Q5 — Turnout (only if time)

- ☐ Scrape 2026 turnout per AC from ECI portal
- ☐ Save `data/external/turnout_2026.csv` with source URL + scrape date
- ☐ Merge; compute delta vs 2021; top 20 increases
- ☐ If skipped: one limitations bullet in deck

PHASE 3 — Visual design system (Day 5)

- ☐ **Palette:** Colorblind-safe (e.g. Okabe-Ito); avoid saffron/green combos that imply parties
- ☐ **Fonts:** Single sans-serif; large axis labels for TV
- ☐ **Titles:** Fact-only — “Average winning margin fell from 11.8% to 7.7%” not “DMK crushed”
- ☐ Export charts at 1920×1080 min → `outputs/charts/`
- ☐ Chart inventory table:

Chart ID	Question	File	Deck slide
C1	Q6	margin_hist.png	3
C2	Q2	sankey_flips.png	4
C3	Q3	tvk_by_region.png	5
C4	Q1	seats_by_region.png	6

PHASE 4 — Deck (Days 6–7)

18 slides (current build) — generated by `python scripts/build_presentation.py` :

1. Title — Headline finding
 2. The challenge — RPC #21 problem statement
 3. Our solution — Pipeline → three narratives → dashboard
 4. Technology stack
 5. Data & methodology
 6. Research questions — coverage map (Q1–Q6) + electoral arithmetic note
 7. **Q6 LEAD** — Margin of victory
 8. **Q2 LEAD** — Seat churn at scale
 9. **Q3 LEAD** — Vote share statewide
 0. Q3 — Vote share by region
 1. Q1 — Geographic pattern
 2. Q4 — Reserved seats (GEN / SC / ST)
 3. Q5 — Turnout
 4. **Deep insight** — Electoral arithmetic (ENP / Pedersen / Gallagher)
 5. **Deep insight** — Regional swing heatmap
 6. **Deep insight** — Representation gap (vote vs seat share)
 7. Editorial recommendation — 60-minute show
 8. Limitations & reproducibility
- ☐ Export PDF
 - ☐ Place in `deck/` and link from README

PHASE 5 — Video (Days 8–9)

Step 5.1 — Script (600–900 words for 5–7 min)

Structure:

1. **Hook (30s)**: Headline + one surprising number
 2. **Context (45s)**: AtliQ brief — neutral facts only
 3. **Story 1 (90s)**: Margins — show C1
 4. **Story 2 (90s)**: Flips — show C2
 5. **Story 3 (90s)**: TVK / regions — show C3–C4
 6. **Recommendation (60s)**: Show segment plan
 7. **Limitations (30s)**: Honest caveats
 8. **Close (15s)**: Thank you
- ☐ Bullet script, not word-for-word essay — avoids “reading script” penalty

- ☐ Rehearse 3 times; record face + screen share

Step 5.2 — Production

- ☐ External mic; quiet room
 - ☐ 1080p screen recording; charts legible
 - ☐ Upload YouTube **unlisted** or Google Drive public link
-

PHASE 6 — Dashboard (Days 9–10, optional)

If built — **Streamlit recommended** for speed:

- ☐ Page 1: Filters (region, reserved, flip yes/no)
 - ☐ KPI cards: total flips, TVK seats, avg margin 2026
 - ☐ Table: AC list sortable by margin
 - ☐ Do not duplicate every deck chart — exploration only
-

PHASE 7 — QA & submission (Days 10–12)

Step 7.1 — Reproducibility

- ☐ Fresh clone → `pip install -r requirements.txt` → run notebooks top-to-bottom
- ☐ All output numbers match deck/video (spreadsheet cross-check)

Step 7.2 — Tone audit

For every slide title and chart title, ask:

"Would a supporter of any party find this offensive or partisan?"

- ☐ Replace loaded words: "wave", "sweep", "rebel", "vs battleground"

Step 7.3 — Reviewer simulation

- ☐ Watch video once without deck — write one-sentence headline
- ☐ If unclear, re-cut opening 60 seconds

Step 7.4 — LinkedIn post

Include:

- ☐ Link to GitHub
- ☐ Link to video
- ☐ Tags: `@codebasics`, Hemanand Vadivel, Dhaval Patel, Government of Tamil Nadu
- ☐ Hashtags: `#ResumeProjectChallenge #Codebasics #DataAnalytics #TamilNaduElection2026`

Step 7.5 — RPC portal

- ☐ Submit LinkedIn post URL on Resume Project Challenge page before deadline
-

8. Evaluation rubric map (100 points)

Criterion	Pts	How this plan addresses it
Story selection & framing	20	Single headline; 3 linked questions; regional slide
Video performance	20	Script structure; rehearsal; hook + recommendation
Deck narrative arc	15	8-slide arc with editorial rec
Data accuracy & reproducibility	20	Metric doc, validation gates, public GitHub
Dashboard	10	Optional Streamlit
Non-partisanship	10	Tone audit checklist
Limitations honesty	5	Dedicated slide + video beat

9. Timeline (working backward from 28 May 2026)

Dates	Milestone
Day 1–2	Repo + AC tables + party normalization
Day 3–4	All core metrics + charts C1–C4
Day 5	Visual polish + chart exports
Day 6–7	Deck v1 → final PDF
Day 8–9	Video script, record, upload
Day 10	Dashboard (optional) + QA
Day 11–12	Tone audit, LinkedIn, RPC submit
28 May	Buffer / only fixes

10. Risk register

Risk	Mitigation
Party label chaos in 2026	Normalization table + document uncertainty
163 raw flips include alliance artifacts	Report normalized flips; footnote methodology
Turnout blank in 2026 CSV	Skip Q5 or scrape early with dated external file
Constituency name mismatch	Join only on <code>ac_number</code>
Over-claiming TVK "vote source"	Descriptive share changes only
Disqualification on tone	Tone audit + no causal language

11. Quick reference — file paths in workspace

```
input_files_for_participants_rpc/
├── Readme-first-brief.docx
├── research_questions_and_recommendations.docx
├── How_to_Submit.docx
├── metadata.txt
├── data/
│   ├── tn_2021_results.csv
│   ├── tn_2026_results.csv
│   └── constituency_master.csv
```

12. Decision log (update as you go)

Date	Decision	Rationale
	Headline text	
	3 questions chosen	Q6, Q2, Q3 recommended
	Turnout: include Y/N	
	Dashboard tool	

Last updated: 28 May 2026 — added deep electoral-arithmetic layer (ENP, Pedersen, Gallagher), new `/deep` page, 18-slide deck.